

Using only two magnetorquers to de-tumble a 2U CubeSAT

Matthew Monkell¹, Carlos Montalvo^{*,2}, Edmund Spencer³

Facility for Aerospace Systems and Technology (FAST), College of Engineering, University of South Alabama, 150 Jaguar Dr., Shelby Hall Rm. 3125, Mobile, AL 36688, USA

Received 24 March 2018; received in revised form 15 August 2018; accepted 27 August 2018
Available online 3 September 2018

Abstract

A CubeSAT is a small satellite on the order of 10 cm along each axis. A 1U satellite is a small cube with 10 cm sides. A 2U CubeSAT has the volume of two single 1U satellites. The size of the satellite is $10 \times 10 \times 20$ cm. These satellites are used for a variety of missions and created by a variety of different organizations. When deployed from a rocket, a CubeSAT may obtain a large angular velocity which must be reduced before most science missions or communications can take place. Maximizing solar energy charging also involves better pointing accuracy. To control the attitude of these small satellites, thrusters, reaction wheels or magnetorquers are used. On a standard CubeSAT, 3 reaction wheels are used as well as 3 magnetorquers. In the initial phase of the CubeSAT mission, the magnetorquers are used to reduce the angular velocity of the satellite down to a manageable level. Once the norm of the angular velocity is low enough, the reaction wheels can spin up reducing the angular velocity to zero. This paper derives a simple control scheme that allows the performance of the de-tumbling maneuver to decline while reducing the number magnetorquers by 1. Thus, the de-tumbling maneuver is completed using 2 magnetorquers rather than 3. This reduces complexity of the satellite, saves weight and reduces energy consumed by the satellite which can be used for other power hungry devices.

Published by Elsevier Ltd on behalf of COSPAR.

Keywords: Cubesats; Magnetorquers; Satellite dynamics; Attitude dynamics

1. Introduction

CubeSats, small satellites found in low or medium Earth orbit (LEO/MEO), typically collect data such as atmospheric, radiation, and geographic readings. These small satellites range in size with a standard unit of one liter (1U). The mass of these 1U CubeSats are on the order of 1 kg (Ratcliffe and Montalvo, 2015). These satellites were originally designed to set a standard for future picosatellites. Since these satellites have such a comparatively low

price tag, there has been an increase in demand for new technologies and research. Multiple universities and institutes around the world have sent their own CubeSats into space for development, research, earth science, and other uses. For example, Auburn University's CubeSat, AubieSat-1, tested how well different solar panels endured against the environment as well as transmitting Auburn University's motto, War Eagle back to Earth in Morse code (Wersinger, 2015). CubeSats were not originally designed to a set standard. The concept began in 1999 as a design challenge for graduate students, but now the program is building on 18 years of direct experience and 60 years of small satellite technology (Kramer and Cracknell, 2008). There have been 341 missions identified by Saint Louis University including the first CubeSat sent into space by the military just a year after the CubeSat concept was proposed. PicoSAT 1 and 2, weighing 250 grams

* Corresponding author.

E-mail addresses: mem1421@jagmail.southalabama.edu (M. Monkell), cmontalvo@southalabama.edu (C. Montalvo), espencher@southalabama.edu (E. Spencer).

¹ Research Engineer, Department of Mechanical Engineering.

² Assistant Professor, Department of Mechanical Engineering.

³ Associate Professor, Department of Electrical Engineering.

Nomenclature

x, y, z	components of the mass center position vector in the inertial frame (m)	m	mass (kg)
ϕ, θ, ψ	Euler roll, pitch, and yaw angles (rad)	I	mass moment inertia matrix about the mass center in the body frame (kg m ²)
q_0, q_1, q_2, q_3	quaternions	X, Y, Z	components of the total force applied to CubeSAT in body frame (N)
u, v, w	components of the mass center velocity vector in the body frame (m/s)	L, M, N	components of the total moment applied to CubeSAT in body frame (N m)
p, q, r	components of the mass center angular velocity vector in the body frame (rad/s)	$\vec{r}_{A \rightarrow B}$	position vector from a generic point A to a generic point B (m)
$\vec{\omega}_{B/I}$	angular velocity vector of the satellite in the body frame (rad/s)	$\vec{v}_{A/B}$	velocity vector of a generic point A with respect to a generic frame B (m/s)
T_{IB}	rotation matrix from frame I to frame B	$S(\vec{r})$	skew symmetric matrix operator on a vector. Multiplying this matrix by a vector is equivalent to a cross product
H	relationship between angular velocity components in body frame and derivative of Euler angles		

each, were launched into orbit from the rocket Minotaur-1 on February 6, 2000. Government agencies such as the National Aeronautics and Space Administration (NASA) as well as the European Space Agency (ESA) are also interested in these picosatellites, assisting various team's CubeSats into orbit around Earth (Mahoney, 2013).

Attitude determination and control systems (ADCS) are required for all space missions (Najam Abbas et al., 2012). The concept of controlling a spacecraft is fairly standard. The main control mechanisms employed on spacecraft include thrusters, magnetorquers (Celani, 2015) and reaction wheels (Najam Abbas et al., 2012). However, some other common ways the motion of a picosatellite is controlled is using whip antennas, deployable arrays and tethers. Since CubeSats are relatively new, there is no common qualification standard and these mechanisms are largely custom made for a single mission (Oppenheimer, 2009). Given the small size of CubeSats and relatively low budgets, thrusters are not typically found on CubeSats. Although they do exist, most universities simply use reaction wheels and magnetorquers. When a CubeSAT is initially deployed, magnetorquers are typically used to reduce the angular velocity vector down to a manageable level such that reaction wheels can take over without saturation. The reaction wheels are then desaturated using magnetorquers again (Giulietti et al., 2006). Typical CubeSats use three magnetorquers combined with three reactions wheels. Much work has been performed on attitude stabilization of CubeSats using magnetorquers (Giulietti et al., 2006; Celani, 2015; Najam Abbas et al., 2012; Lin and Lee, 2015; Gulmammadov et al., 2010). Attitude stabilization using magnetorquers is accomplished using standard Bdot controllers which involves computing the derivative of the magnetic field and applying control opposing that change (Assaad, 2013).

All previous work has relied on using three magnetorquers oriented along the three principal axes of inertia of the satellite. The work presented here derives a controller

that seeks to de-tumble the satellite while only using two magnetorquers. Although performance is reduced, it would be possible to use this controller in the event a magnetorquer fails or can be used to cut costs in hardware. The sections that follow detail the dynamic model used to simulate the orientation of the satellite as well as the control system model for various types of controllers. Simulation results are shown to highlight the advantages and disadvantages of using two rather than three magnetorquers.

2. Mathematical model

The mathematical model used here begins with the Earth Magnetic Field and Gravitational Models. The magnetic field model comes from the Geographic Library model which uses the EMM2015 magnetic field model. The gravitational model comes from the EGM2008 model (Geographic, 2017). Then the translational equations of motion of the satellite are written using inertial coordinates for simplicity. The satellite is assumed to be a rigid body using quaternions to parameterize orientation. Magnetorquers are modeled as well to test controllers assuming a certain amount of current and number of turns in the coils.

2.1. Magnetic Field Model

The Magnetic Field model used in this simulation stems from the Enhanced Magnetic Field Model (EMM2015) (Enhanced, 2017). The Earth's magnetic is a complex superposition of multiple sources including the inner core and outer core of the planet. Models have been created that use spherical harmonics to compute the magnetic field at any location around the Earth. The EMM2015 model uses a 720 order model increasing the spatial resolution down to 56 km. This model was compiled from multiple sources including but not limited to satellite and marine data. It also includes data from the European Space Agency's Swarm satellite mission. In order to include this harmonic

mesh data into this simulation, the GeographicLib module written in C++ is included in the simulation (Geographic, 2017). The result is the ability to give any position coordinate of the satellite and have the model return the magnetic field strength in spherical coordinates. The inputs to the model are the position x, y, z of the satellite assuming an inertial frame with the z -axis pointing through the north pole and the x axis pointing through the equator at the prime meridian as seen in Fig. 1. This is also known as the Earth-Centered Inertial (ECI) coordinate system (Ashby, 2004). In order to connect these inertial coordinates (x, y, z) to be used in the EMM2015 model, the latitude, longitude and height are required. To do this, the coordinates are converted into spherical coordinates using the equations below.

$$\begin{aligned}\rho &= \sqrt{x^2 + y^2 + z^2} \\ \phi_E &= 0 \\ \theta_E &= \cos^{-1} \left(\frac{z}{\rho} \right) \\ \psi_E &= \tan^{-1} \left(\frac{y}{x} \right)\end{aligned}\quad (1)$$

Note that $\rho, \phi_E, \theta_E, \psi_E$ are related to latitude and longitude coordinates but not quite the same. In order to obtain the latitude and longitude coordinates the following equations are used. The height is simply the distance from the center of the ECI frame minus the reference height from the approximation of Earth as an ellipsoid. Note that the angles from Eq. (1) are converted to degrees.

$$\begin{aligned}\lambda_{LAT} &= 90 - \theta_E \frac{180}{\pi} \\ \lambda_{LON} &= \psi_E \frac{180}{\pi}\end{aligned}\quad (2)$$

The inputs are then the latitude, longitude and height. The output is the north, east, and up component in the spherical reference frame. The spherical coordinates of the magnetic field obtained from the EMM2015 model can then be rotated to the inertial frame using the following equation where $\vec{\beta}_{SPH}$ is the magnetic field in the spherical coordinate system and $\vec{\beta}_I$ is the magnetic field in the inertial frame.

$$\vec{\beta}_I = \mathbf{T}(\phi_E, \theta_E, \psi_E) \vec{\beta}_{SPH} \quad (3)$$

The matrix $\mathbf{T}(\phi, \theta, \psi)$ represents the transformation matrix from the spherical reference frame to the inertial reference frame, which is constructed using the standard aircraft flight mechanics rotation sequence (3–2–1). Standard shorthand notation is used for trigonometric functions: $\cos(\alpha) \equiv c_\alpha$, $\sin(\alpha) \equiv s_\alpha$, and $\tan(\alpha) \equiv t_\alpha$.

$$\mathbf{T}(\phi, \theta, \psi) = \begin{bmatrix} c_\theta c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix} \quad (4)$$

The result of these equations, is the ability to obtain the magnetic field across an entire orbit. Fig. 2 shows an example 56 degree inclination orbit, 600 km above the Earth's surface. The orbit begins with the satellite above the equator and the prime meridian and assumes the Earth does not rotate.

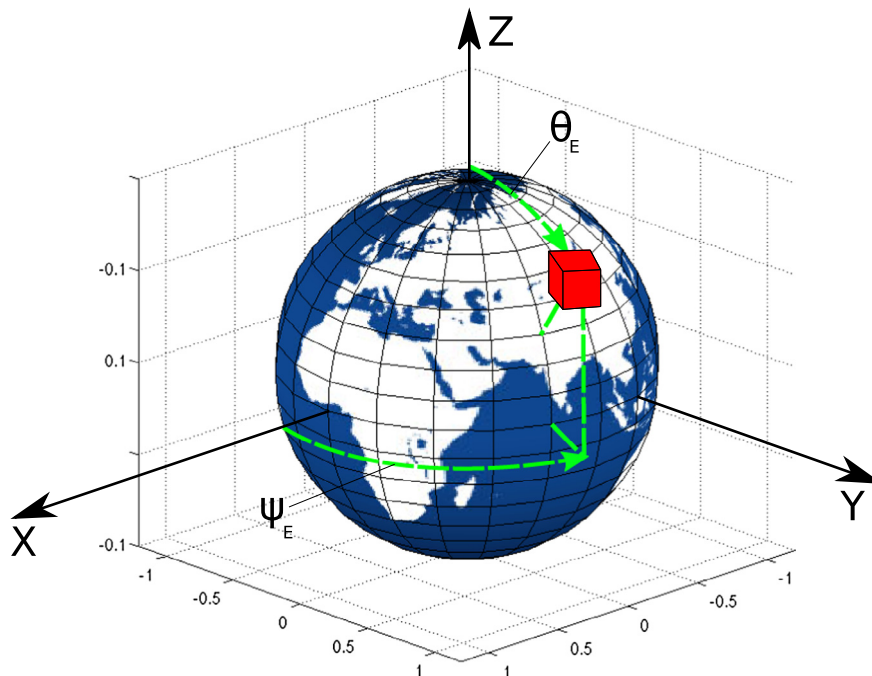


Fig. 1. Earth-centered inertial frame and spherical coordinate frame.

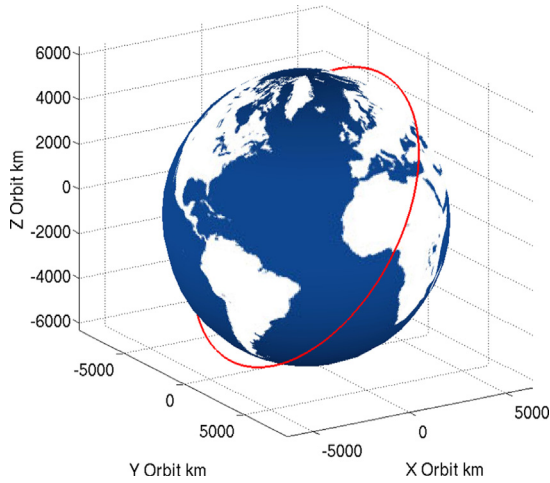


Fig. 2. Example 56 Degree inclination orbit at 600 km above Earth's surface.

Fig. 3 shows the magnetic field during the orbit in the inertial frame. The x-axis has been non-dimensionalized to represent the entire orbit. Thus when the x-axis is equal to 100 the satellite has completed one orbit.

2.2. Gravitational model

The gravitational field model used in this simulation stems from the Earth Gravity Model (EGM2008) (Earth, 2017) which can also be found in the GeographicLib module (Geographic, 2017). This model computes Earth's gravitational field at any point in three dimensional space. The model takes in coordinates in the ECI frame and returns the gravitational acceleration in the ECI frame thus no rotation is required. Just like the EMM2015 model this model uses spherical harmonics and a reference ellipsoid. The reference ellipsoid is then updated with gravity distur-

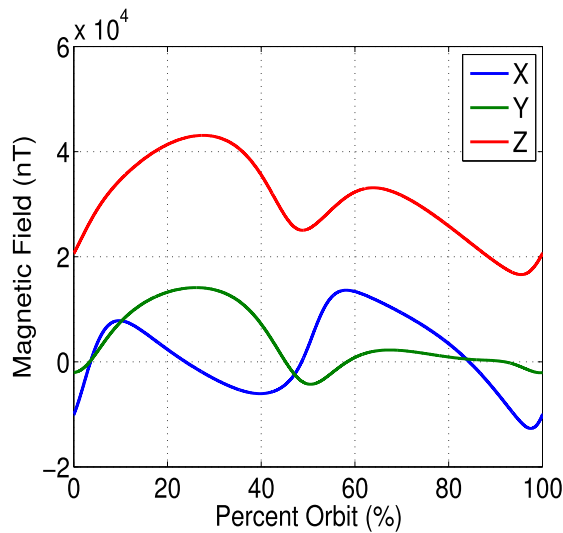


Fig. 3. Magnetic field of earth in inertial frame for 56 degree orbit at 600 km above surface.

bances such as non-uniform geoid heights. This model is an upgrade from EGM84 and EGM96 which only used models of order 180 and 360 respectively. The EGM2008 model as a comparison uses a model of order 2190. Fig. 4 shows the gravitational acceleration vector during a 56 degree orbit at 600 km above the Earth's surface. The x-axis has been non-dimensionalized to represent the entire orbit. Thus when the x-axis is equal to 100 the satellite has completed one orbit.

2.3. Translational equations of motion

The translational equations of motion of the satellite are fairly simple given that everything is written in the inertial frame. The position vector of the satellite is $\vec{r} = [x, y, z]^T$ and the velocity is $\vec{V}_{B/I} = [\dot{x}, \dot{y}, \dot{z}]^T$. The acceleration of the satellite is found by summing the total forces on the body and dividing by the mass of the satellite. Since the only external force on the satellite is the acceleration due to gravity, the equation breaks down to a quite trivial result.

$$\vec{a}_{B/I} = \vec{a}_G \quad (5)$$

Note that the magnitude of the gravitational acceleration vector is on the order of $\pm 10 \text{ m/s}^2$. Other forces such as solar radiation pressure and atmospheric drag are neglected due to their small force as compared to gravity. Sources point to solar radiation pressure being on the order of $4.5 \text{ } \mu\text{Pa}$ (Georgevic, 1973). For a 1U CubeSat ($10 \text{ cm} \times 10 \text{ cm}$) the force would be equal to 0.45 mN . A 1U CubeSat has a nominal mass of 1 kg which would accelerate the CubeSat on the order of 0.45 mm/s^2 , which is considerably less than gravitational acceleration. Furthermore, using the standard aerodynamic drag equation ($0.5\rho V^2 SC_D$), where conservative estimates are used, the aerodynamic force at 600 km above the Earth's surface would be about 3.0 nN (Anderson, 2007). This assumes a density equal to $1.03 \times 10^{-14} \text{ kg/m}^3$, a velocity equal to 7.56 km/s , and a drag coefficient equal to 1.0 (Gyatt, 2006). A force this small would impart an acceleration of about 3.0 nm/s^2 which is also considerably less than gravitational acceleration. Although these forces cannot be neglected for longer missions, they are neglected here since only a few orbits are simulated.

2.4. Attitude equations of motion

The attitude equations of motion are formulated assuming the satellite can rotate about three axes. It is well known that equations of motion produced by using only three orientation parameters results in a singularity (Etkins, 2000). As such, the orientation of the satellite is parameterized using four parameters known as quaternions (Lin and Lee, 2015; Filipe and Tsiotras, 2015). The kinematic equations for quaternions are given by Eq. (6).

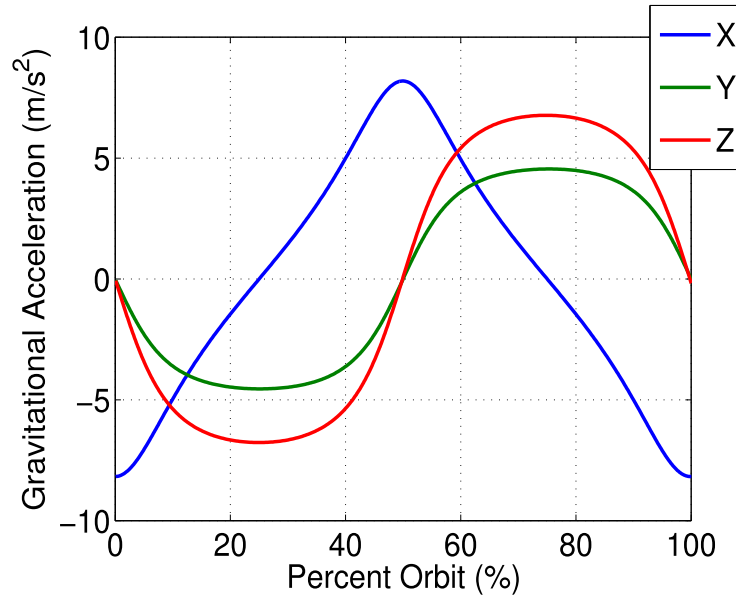


Fig. 4. Gravitational field of earth in inertial frame for 56 degree orbit at 600 km above surface.

$$\begin{Bmatrix} \dot{q}_0 \\ \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{Bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & -p & -q & -r \\ p & 0 & r & -q \\ q & -r & 0 & p \\ r & q & -p & 0 \end{bmatrix} \begin{Bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{Bmatrix} \quad (6)$$

where q_i are the four quaternions and p, q, r are the components of the angular velocity vector in the body frame. The derivative of angular velocity is found by equating the derivative of angular momentum to the total moments placed on the satellite (Etkins, 2000).

$$\begin{Bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{Bmatrix} = [I]^{-1} \left(\begin{Bmatrix} L \\ M \\ N \end{Bmatrix} - \mathbf{S}(\vec{\omega}_{B/I})[I] \begin{Bmatrix} p \\ q \\ r \end{Bmatrix} \right) \quad (7)$$

The skew symmetric operator is denoted by $\mathbf{S}()$. Multiplying this matrix by a vector is equivalent to a cross product.

$$\mathbf{S}(\vec{\omega}_{B/I}) = \begin{bmatrix} 0 & -r & q \\ r & 0 & -p \\ -q & p & 0 \end{bmatrix} \quad (8)$$

The mass moment of inertia matrix I is taken about the center of mass of the spacecraft and is positive definite and symmetric. The applied moments expressed in Eq. (7) solely contain contributions from the Magnetorquers. All gravity gradient torques and aerodynamic torques are neglected. For a 1U CubeSat, the gravity gradient over 10 cm is about $0.24 \mu\text{m/s}^2$ using the EGM2008 model. Multiplying this acceleration by a 1 kg mass and applying a 10 cm moment arm yields a moment of about $2.4 \times 10^{-8} \text{ N m}$. Aerodynamic torques could be as large as $1.5 \times 10^{-10} \text{ N m}$ assuming the aerodynamic center is 5 cm away from the center of mass. The magnetorquers oper-

ate in the vicinity of $3.0 \times 10^{-6} \text{ N m}$, assuming a current of 0.04 A, an area of 0.02 m^2 , 84 turns and a magnetic field of 40,000 nT. Using these calculations, the magnetorquers are two orders of magnitude larger than gravity torques and four orders of magnitude larger than aerodynamic torques. This is using fairly conservative estimates as well. In practice, the gravity gradient will not have a moment arm equal to 10 cm Snadden et al., 1998.

2.5. Magnetorquer model

The magnetorquer model assumes that three magnetorquers are aligned in such a way that the magnetic moment produced by each magnetorquer is aligned with the principal axes of the body frame of the satellite. Each magnetorquer is controlled independently such that $\vec{i}_B = [i_x, i_y, i_z]^T$ which is the applied current in each magnetorquer. The magnetic moment is then given by the equation below

$$\vec{\mu}_B = nA \vec{i}_B \quad (9)$$

where n is the number of turns in the coil of each magnetorquer and A is the area of the magnetorquer. For simplicity it is assumed that all magnetorquers have the same area and same number of turns. The torque produced by all magnetorquers is then simply found by crossing the magnetic moment with the magnetic field of the Earth in the Body reference frame.

$$\vec{\tau}_B = \mathbf{S}(\vec{\mu}_B) \mathbf{T}(\vec{q}) \vec{\beta}_I \quad (10)$$

In order to obtain the magnetic field vector in the body frame, the inertial magnetic field vector must be rotated into the body frame of the satellite. This is accomplished by using the quaternion transformation matrix as shown below.

$$\mathbf{T}(\vec{q}) = \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 + q_0q_3) & 2(q_1q_3 - q_0q_2) \\ 2(q_1q_2 - q_0q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_0q_1 + q_2q_3) \\ 2(q_0q_2 + q_1q_3) & 2(q_2q_3 - q_0q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix} \quad (11)$$

In component form, Eq. (10) reduces to the following equation using the identity that $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$

$$\begin{Bmatrix} L \\ M \\ N \end{Bmatrix} = nA \begin{bmatrix} 0 & \beta_z & -\beta_y \\ -\beta_z & 0 & \beta_x \\ \beta_y & -\beta_x & 0 \end{bmatrix} \begin{Bmatrix} i_x \\ i_y \\ i_z \end{Bmatrix} \quad (12)$$

where $\beta_x, \beta_y, \beta_z$ are the components of the magnetic field in the body frame of the satellite. The moments L, M, N are thus the control torques that rotate the satellite as seen in Eq. (7).

2.6. Control system

Two controllers are presented here. The first is the standard B-dot controller reported in many sources (Leomanni, 2012; Lovera, 2015; Assaad, 2013; Sanyal and Lee-Ho, 2009). The second has been coined the Reduced-Row-Echelon-Form (RREF) Controller. Performance from an analytical perspective is shown for both.

2.6.1. B-dot controller

The standard B-dot controller requires the magnetorquers to follow the simple control law shown below

$$\vec{\mu}_B = k\mathbf{S}(\vec{\omega}_{B/I})\mathbf{T}(\vec{q})\vec{\beta}_I \quad (13)$$

where k is the control gain. Using Eq. (9) it is possible to write the current in component form again using the identity that $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$

$$\begin{Bmatrix} i_x \\ i_y \\ i_z \end{Bmatrix} = \frac{k}{nA} \begin{bmatrix} 0 & \beta_z & -\beta_y \\ -\beta_z & 0 & \beta_x \\ \beta_y & -\beta_x & 0 \end{bmatrix} \begin{Bmatrix} p \\ q \\ r \end{Bmatrix} \quad (14)$$

This equation can then be substituted into Eq. (12) to produce the total torque on the satellite assuming that the magnetorquers can provide the necessary current commanded by Eq. (14).

$$\begin{Bmatrix} L \\ M \\ N \end{Bmatrix} = -k \begin{bmatrix} \beta_y^2 + \beta_z^2 & -\beta_x\beta_y & -\beta_x\beta_z \\ -\beta_x\beta_y & \beta_x^2 + \beta_z^2 & -\beta_y\beta_z \\ -\beta_x\beta_z & -\beta_y\beta_z & \beta_x^2 + \beta_y^2 \end{bmatrix} \begin{Bmatrix} p \\ q \\ r \end{Bmatrix} \quad (15)$$

The goal of the controller here is to drive $\vec{\omega}_{B/I} \rightarrow 0$. The literature will show that is not completely achieved (Celani, 2015). There are multiple explanations for this. For starters, Eq. (10) assumes that the magnetic moment is not co-linear with the magnetic field of the Earth. If it is, the result is zero torque applied to the satellite. Furthermore, Eq. (14) results in zero current if the angular velocity vector of the satellite is co-linear with the magnetic field. Thus, if the magnetic field vector, angular velocity vector

or the magnetic moment vector are co-linear, the torque applied to the satellite will be zero. If a new operator is defined such that

$$\mathbf{W}(\mathbf{T}(\vec{q})\vec{\beta}_I) = \begin{bmatrix} \beta_y^2 + \beta_z^2 & -\beta_x\beta_y & -\beta_x\beta_z \\ -\beta_x\beta_y & \beta_x^2 + \beta_z^2 & -\beta_y\beta_z \\ -\beta_x\beta_z & -\beta_y\beta_z & \beta_x^2 + \beta_y^2 \end{bmatrix} \quad (16)$$

it is easy to see that the torque applied to a satellite is then simply the angular velocity vector multiplied by this transition matrix. If this transition matrix is put into row-reduced-echelon form it is easy to see that the determinant of this matrix is equal to zero (Carlen and Carvalho, 2006).

$$\text{rref}(\mathbf{W}(\mathbf{T}(\vec{q})\vec{\beta}_I)) = \begin{bmatrix} 1 & 0 & -\beta_x/\beta_z \\ 0 & 1 & -\beta_y/\beta_z \\ 0 & 0 & 0 \end{bmatrix} \quad (17)$$

A zero determinant means that there exists a vector $\vec{\omega}_{B/I}$ that will result in zero torque for a given value of the magnetic field. This is typically avoided since the magnetic field of the Earth is time and spatially varying which results in a transition matrix that changes over time due to orientation changes in the satellite as well as changes in the satellite's orbit. However, for low inclination orbits, it's possible for the magnetic field to stay relatively constant with $\beta_x \approx \beta_y \approx 0$. If the satellite is tumbling about the yaw axis such that $p = q = 0$, the yaw torque on the satellite (N) will be zero. Using this simple controller, there is no way to remove the remaining angular velocity from the satellite. Work has been done to assist this issue by taking advantage of the spatially varying and almost periodic earth magnetic field (Celani, 2015).

2.6.2. RREF controller

The B-dot controller assumes that three magnetorquers are used to de-tumble the satellite. A new controller is presented here that only uses two magnetorquers to de-tumble the satellite with similar performance to the controller above. To derive this controller, Eq. (12) is revisited. A control vector is defined such that $\vec{u} = k\vec{\omega}_{B/I}$. If the torque on the satellite $\vec{\tau}_B$ is set equal to this control vector, the satellite will de-tumble and result in zero angular velocity.

$$\frac{k}{nA} \begin{Bmatrix} p \\ q \\ r \end{Bmatrix} = \begin{bmatrix} 0 & \beta_z & -\beta_y \\ -\beta_z & 0 & \beta_x \\ \beta_y & -\beta_x & 0 \end{bmatrix} \begin{Bmatrix} i_x \\ i_y \\ i_z \end{Bmatrix} \quad (18)$$

Solving the equation above for the current $\vec{i}_B$ requires a simple matrix inversion; however, it is well known that the skew symmetric matrix has no inverse with a zero determinant due to the co-linearity of cross-products (Carlen and Carvalho, 2006). To solve this, Gaussian Elimination is performed which results in the following set of equations where $\bar{M}_p = kp/nA$, $\bar{M}_q = kq/nA$ and $\bar{M}_r = kr/nA$.

$$\begin{Bmatrix} \bar{M}_r/\beta_y \\ \bar{M}_p/\beta_z \\ \bar{M}_p\frac{\beta_x}{\beta_y} + \bar{M}_q + \bar{M}_r\frac{\beta_x}{\beta_y} \end{Bmatrix} = \begin{bmatrix} 1 & -\beta_x/\beta_y & 0 \\ 0 & 1 & -\beta_y/\beta_z \\ 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} i_x \\ i_y \\ i_z \end{Bmatrix} \quad (19)$$

As expected, Gaussian Elimination only results in two equations that can be used to solve for current in two magnetorquers.

$$\begin{aligned} i_x &= \bar{M}_r \frac{1}{\beta_y} + \bar{M}_p \frac{\beta_x}{\beta_y \beta_z} + i_z \frac{\beta_x}{\beta_z} \\ i_y &= \bar{M}_p \frac{1}{\beta_z} + i_z \frac{\beta_y}{\beta_z} \end{aligned} \quad (20)$$

Notice that the two equations are functions of i_z . This is a direct result of having an undefined inverse. This represents the fact that if the magnetic field and angular velocity are co-linear the torque drops to zero. However, this can be used as an advantage to reduce the complexity of the satellite. That is, i_z can be set to zero essentially removing the entire third magnetorquer. The result is the following three equations for the current in the magnetorquers.

$$\begin{Bmatrix} i_x \\ i_y \\ i_z \end{Bmatrix} = \frac{k}{nA} \begin{bmatrix} \frac{\beta_x}{\beta_y \beta_z} & 0 & \frac{1}{\beta_y} \\ \frac{1}{\beta_z} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} p \\ q \\ r \end{Bmatrix} \quad (21)$$

Notice, that this controller does not depend on the pitch rate of the satellite which means that satellite will continue to tumble about the pitch axis but the angular velocity in the roll and yaw axes should be completely removed. To investigate this controller analytically, the third equation in Eq. (19) is revisited.

3. Simulation results

For the following simulations, the satellite is modeled after a 2U satellite with a mass of 2.6 kg. The inertia is set to $I_x = I_y = 0.0108 \text{ kg m}^2$ and $I_z = 0.0043 \text{ kg m}^2$. The

magnetorquers have a maximum total current of 0.04 A. That is, the total current supplied to all three or two magnetorquers cannot exceed 0.04 A. The magnetorquer area is 0.02 m^2 with 84 turns per magnetorquer. The B-dot controller has its gain set to -1×10^5 while the RREF controller has its gain set to -1×10^{-4} . The initial angular velocity of the satellite is set to $\vec{\omega}_{B/I} = [0.8, 0.8, 0.8] \text{ (rad/s)}$ which corresponds to about 50 deg/s. The satellite is placed in a 56 degree inclination orbit, 600 km above the Earth's surface. The simulation has a timestep of 0.01 and uses a fixed step runge-kutta-4 integration scheme. The simulation is run for 1000 min or about 16 h. A satellite at 600 km would complete about 10 orbits in that time. Angular rate data is shown for both the Bdot and RREF controllers in Figs. 5–7. The current supplied to the magnetorquers are shown in Fig. 8.

Figs. 5–7 indicate a few things about the performance of the two controllers. The first result is that the Bdot controller de-tumbles the satellite in about 800 min as compared to the RREF controller which takes about 250 min. This large difference in time is attributed to the fact that the RREF controller hits a singularity and no longer slows the satellite. Notice that the roll rate and yaw rate tends to zero in both controllers but the pitch rate reaches a constant in the RREF controller while the Bdot controller completely removes angular velocity. If the magnetic field were constant the Bdot controller would not be able to completely remove the angular velocity but since the magnetic field changes as a function of orbit the Bdot controller can completely remove angular velocity. The RREF controller unfortunately only has two magnetorquers and thus leaves the pitch rate non-zero. The steady state angular velocity is then $\vec{\omega}_{B/I} = [0.0, -0.64, 0.0]^T \text{ rad/s}$. The non-zero value in pitch rate is an expected result as given by Eq. (21). It may be assumed at this point that the RREF controller is inferior to the Bdot controller. However, there is a benefit to having the satellite de-tumble along two axes. Using this RREF controller, it would also be possible to only use 1 reaction wheel since the satellite consistently

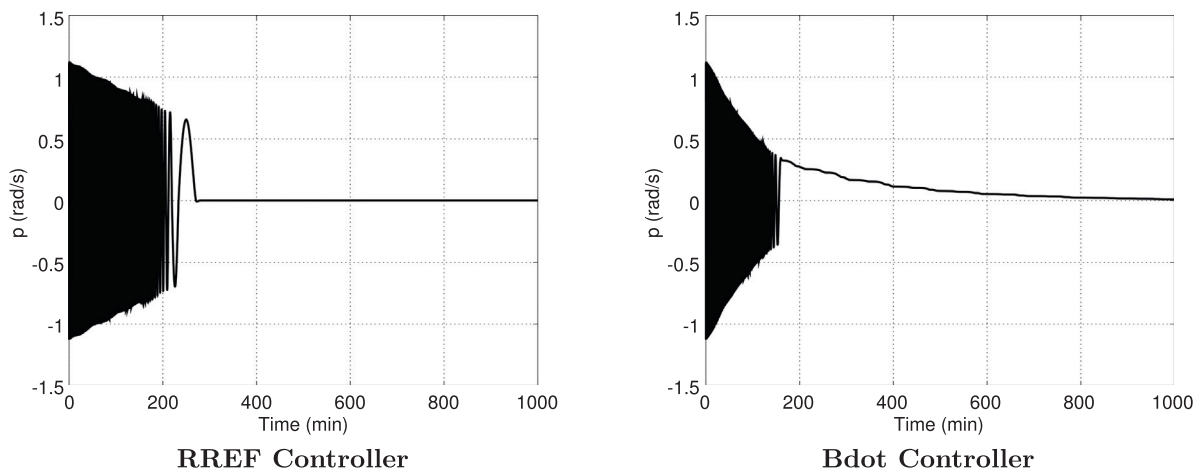


Fig. 5. Roll Rate of Satellite (rad/s) vs. Time (min) for RREF (left) and Bdot (Right) Controllers.

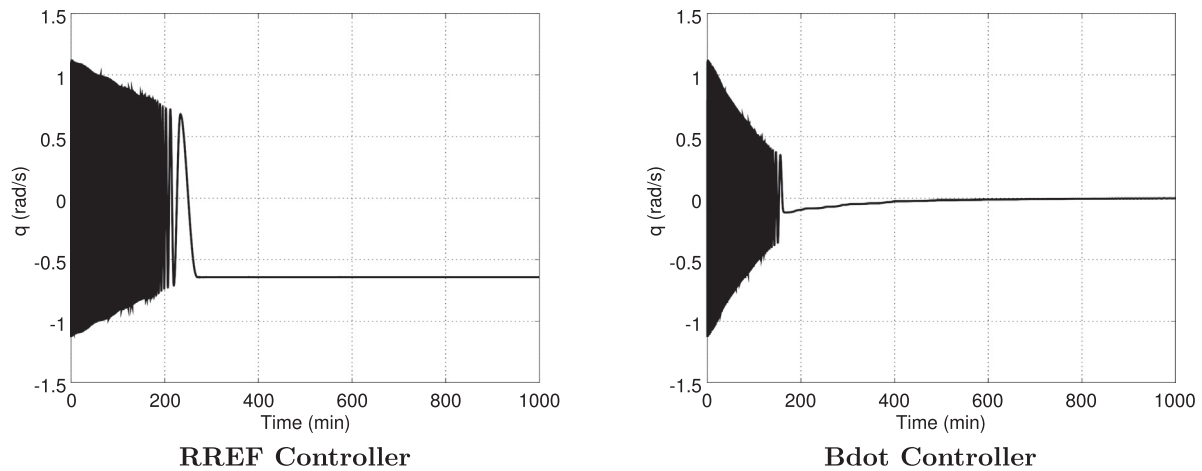


Fig. 6. Pitch rate of satellite (rad/s) vs. time (min) for RREF (left) and Bdot (Right) controllers.

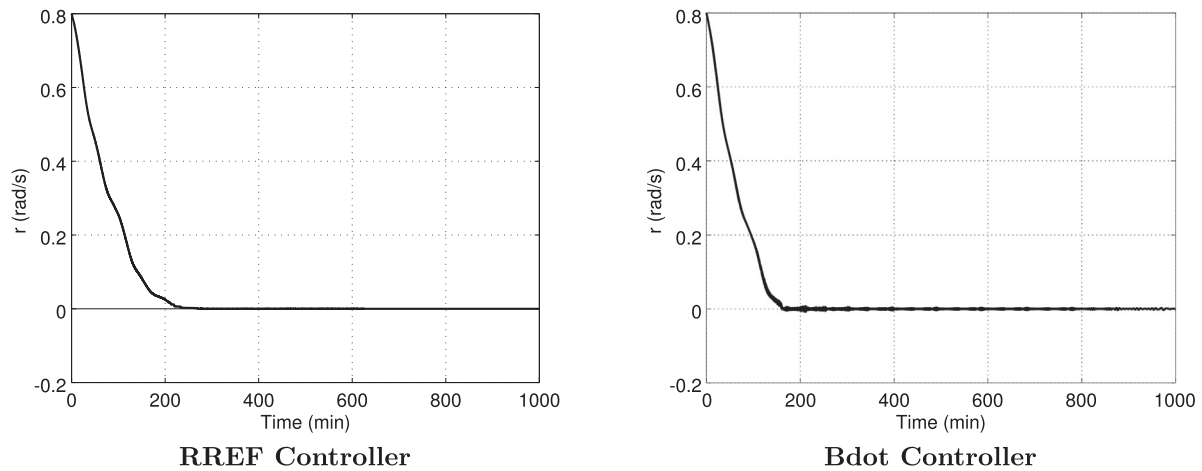


Fig. 7. Yaw rate of satellite (rad/s) vs. time (min) for RREF (left) and Bdot (Right) controllers.

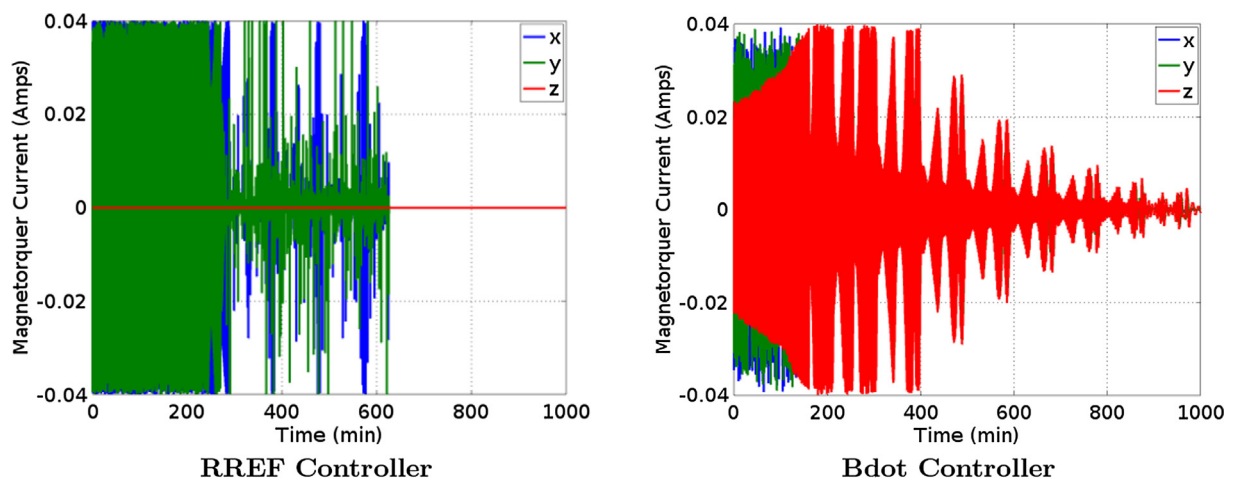


Fig. 8. Magnetorquer current (Amps) vs. time (min) for RREF (left) and Bdot (Right) controllers.

tumbles about the pitch axis, thus removing even more hardware from the satellite. Fig. 8 again highlights the detumbling maneuver happens in about 200 min with a few correction up to about 600 min. The Bdot controller

has an exponential decay but is still consuming power up to 1000 min. The main difference between the two plots is the fact that the z-axis magnetorquer current is zero for the entire duration of the flight when the RREF controller

is used. This again highlights the fact that only two magnetorquers are used which reduces complexity of the system.

4. Conclusions

This paper investigates using only two magnetorquers on a 2U CubeSat with a focus on developing a control law to remove angular velocity from the satellite. Standard practice is to use three magnetorquers and a Bdot controller to remove angular velocity. Typically magnetorquers are actuated until as much angular velocity is removed and then reaction wheels are activated to remove any residual angular velocity. Simulation results are presented for a 2U satellite showing the effectiveness of the Bdot controller which commands current using the cross product of the magnetic field and the angular velocity vector. This results in the satellite reaching a steady state angular velocity vector that is co-linear with the magnetic field. If the orbit of the satellite is such that the magnetic field varies as a function of orbit location, three magnetorquers can completely remove the angular velocity from the satellite. A new controller coined the RREF controller is derived here which assumes the z-axis magnetorquer is set to zero. This results in a steady state angular velocity vector that has roll and yaw rates at zero with a non-zero pitch rate. This new controller reduces the number of magnetorquers by one and also introduces the ability for the satellite to only use one reaction wheel. If weight, volume and cost constraints are a huge design issue, this controller could prove useful for future small satellite missions.

References

- Anderson, D., 2007. *Fundamentals of Aerodynamics* 4th Edition. McGraw Hill Series.
- Ashby, Neil, 2004. The sagnac effect in the global positioning system. *Relativity in rotating frames: relativistic physics in rotating reference frames*, p. 11.
- Assaad, Farhat, 2013. *Attitude Determination and Control System for a CubeSat* (Ph.D. thesis). Worcester Polytechnic Institute.
- Carlen, E., Carvalho, M.C., 2006. *Linear Algebra: From the Beginning*. W.H. Freeman.
- Celani, Fabio, 2015. Robust three-axis attitude stabilization for inertial pointing spacecraft using magnetorquers. *ScienceDirect: Acta Astronautica* 107, 87–96, February–March.
- Earth gravity model. <<https://www.ngdc.noaa.gov/geomag/EMM/>> (cited Jul 31, 2017).
- Enhanced magnetic field model. <<https://www.ngdc.noaa.gov/geomag/EMM/>> (cited Jul 11, 2017).
- Etkins, B., 2000. *Dynamics of Atmospheric Flight*. Dover, Mineola, New York.
- Filipe, Nuno, Tsiotras, Panagiotis, 2015. Adaptive position and attitude-tracking controller for satellite proximity operations using dual quaternions. *J. Guid. Control Dyn.* 38 (4), 566–577.
- Geographic library c++. <<https://geographiclib.sourceforge.io/>> (cited Jul 11, 2017).
- Georgevic, R.M., 1973. The solar radiation pressure forces and torques model. *J. Astronaut. Sci.* 27 (1).
- Giulietti, Fabrizio, Quarta, Alessandro A., Tortora, Paolo, 2006. Optimal control laws for momentum-wheel desaturation using magnetorquers. *J. Guid. Control Dyn.* 29 (6), 1464–1468.
- Gulmammadov, F., Kahraman, O., Yavuzylmaz, C., Tufekci, C.S., Subasi, Y., 2010. Magnetorquers only attitude maintaining using dynamic attitude simulator environment. In: *AIAA Modeling and Simulation Technologies Conference*, Toronto, Ontario Canada.
- Gyatt, Graham, 2006. The standard atmosphere, A mathematical model of the 1976 U.S. Standard Atmosphere.
- Kramer, H.J., Cracknell, A.P., 2008. An overview of small satellites in remote sensing. *Int. J. Remote Sens.* 29 (15), 4285–4337.
- Leomanni, Mirko, 2012. Comparison of control laws for magnetic detumbling. *Research Gate* (accessed: 2/9/2018).
- Lin, Arthur K.L., Lee, Regina, 2015. Attitude control for small spacecraft with sensor errors. In: *AIAA SPACE Conferences and Exposition*, Pasadena, California, p. 4423.
- Lovera, M., 2015. Magnetic satellite detumbling: the b-dot algorithm revisited. In: *2015 American Control Conference (ACC)*, pp. 1867–1872.
- Mahoney, E., 2013. Cubesats initiative. <http://www.nasa.gov/directorates/heo/home/CubeSats_initiative.html> (accessed: 17-Jun-2015).
- Abbas, Naqvi Najam, Xiao, Han, Jun, Li Yan, Raza, Muhammad, 2012. An architecture analysis of adcs for cubesat: a recipe for adcs design of icube. In: *Mechanical and Aerospace Engineering, ICMAE2011. Applied Mechanics and Materials*, vol. 110. Trans Tech Publications, pp. 5397–5404.
- Oppenheimer, Paul, 2009. Cubesat mechanisms workgroup. In: *6th Annual CubeSat Developers Workshop* Cal Poly State University, San Luis Obispo, California, April 2009.
- Ratcliffe, Alicia, Montalvo, Carlos, 2015. Orbital mechanic simulation of a 1u cubesat. In: *Proceedings of the 15th Early Career Technical Conference 2015*, Birmingham, Alabama, USA, November 2015.
- Sanyal, Amit, Lee-Ho, Zachary, 2009. Attitude tracking control of a small satellite in low earth orbit.
- Snadden, M.J., McGuirk, J.M., Bouyer, P., Haritos, K.G., Kasevich, M. A., 1998. Measurement of the earth's gravity gradient with an atom interferometer-based gravity gradiometer. *Phys. Rev. Lett.* 81 (Aug), 971–974.
- Wersinger, J.-M., 2015. Auburn university student space program auburn university student space program. <<http://www.space.auburn.edu/>> (accessed: 17-Jun-2015).